

FromDonna Persistence Plan

Per-turn R2 backup of Hermes brain + workspace

For team review · Implementation plan · July 2026

One-liner (Slack / email)

*We're adding **per-turn R2 backup** of each user's Hermes home + workspace (no secrets), async after each successful Telegram turn, with **restore on sandbox create**— E2B pause stays for idle; R2 is the durable safety net so users don't lose state if the box is replaced.*

1. Brief for team review

What we're building

Per-turn durable backup of each user's E2B sandbox state to Cloudflare R2, plus restore when a sandbox is created or replaced—so users feel continuity after idle pause *and* after hard loss (kill / rebuild), without putting secrets on R2.

Why

Today, continuity for normal use depends on **E2B pause/resume** (good for inactivity). There is **no R2 backup/restore** in code. If a sandbox is lost or re-provisioned, the Hermes brain, chat/session DB, and workspace files die with the box. Product docs already describe R2; we haven't implemented it.

What gets saved (after each successful turn)

- **/home/user/hermes** — agent brain (config, skills, memories, state.db / sessions), filtered
- **/home/user/workspace** — agent working files
- **Never:** user key files / secrets (.env, auth.json, capability tokens, etc.)
- **Skip:** caches, venvs, install tree (/opt/fromdonna), runtime PIDs

How it fits the current system

User message → Gateway Worker → E2B connect/resume → harness /turn → Hermes ↓ async: package filtered state → R2 (user already got Telegram reply)

- **Idle:** E2B auto-pause/resume stays as-is (unchanged).
- **Per turn:** Worker writes to R2 under users/{userId}/... after a successful turn.
- **New/replaced sandbox:** restore latest R2 snapshot before serving turns.
- **Product file tools → R2:** later/same program (architecture path C); not blocking v1 of brain+workspace save.

What we are not changing

- Secrets still only on Worker / Nango / LLM proxy
- No long-lived R2 keys in the sandbox if we can avoid it
- Template contents / Hermes install path stay shared, not per-user R2

Risks to call out

- Per-turn full dumps can get large/slow → start full, then incremental
- Save should run **after** the user reply (async) so chat stays snappy

- Oneshot session resume (--continue fromdonna) may need a small Hermes/harness fix so chat history matches restored state.db
- Failed re-provision today creates an empty new box—restore on create is required for the promise to hold

Success criteria

- After N successful turns, R2 has a current brain + workspace for that user.
- Kill/recreate sandbox → restore → user sees same memory, sessions, and workspace as after last successful turn.
- Normal pause/resume still works with no extra user-facing steps.
- No secrets in R2.

2. Full plan (implementers)

Goals

Goal	Approach
User feels “nothing changed” after idle	Keep E2B pause/resume as primary live continuity
Survive kill / rebuild / lost sandbox	Per-turn save of brain + workspace + sessions to R2; restore on provision
Product files durable	Path C: filesystem sync now; tool→Worker→R2 as follow-on
No secret leakage	Exclude key files; prefer Worker-side R2 credentials

Non-goals (v1)

- Replacing E2B pause with R2 as live filesystem
- Backing up the entire VM / OS packages
- Syncing on every tool call inside a turn
- Putting R2 as the agent's primary working filesystem mid-turn

R2 key layout

```
users/{userId}/ hermes/ # brain archive or object tree workspace/ # workspace archive or object tree
manifests/last-turn.json # turn metadata, hashes, sandbox_id
```

last-turn.json fields (example): `userId`, `sandboxId`, `savedAt`, `turnId` / message id, content hashes, schema version, sizes.

Preferred flow: Worker-orchestrated save (safer)

1. Turn succeeds (POST /turn → { text }).
2. Worker sends Telegram reply.
3. `waitUntil`: package filtered paths from sandbox (E2B command or FS API) → upload R2 → write manifest.
4. Soft-fail save (log/metric); don't fail the user reply if R2 blips.

Alternate: short-lived presigned URLs into the sandbox for push upload (only if archives are large / Worker limits hurt).

Restore flow (on create / failed recovery)

1. Create sandbox → health → bootstrap (unchanged).
2. If `manifests/last-turn.json` exists for `userId` → restore hermes + workspace.
3. Mark ready → accept turns.

Include / exclude

Include	Exclude
<code>~/hermes/**</code> (filtered; align with Hermes backup.py)	<code>.env</code> , <code>auth.json</code> , <code>tokens/keys</code>
Consistent state.db snapshot (SQLite backup API)	<code>node_modules</code> , <code>.venv</code> , <code>__pycache__</code> , <code>.cache</code> , <code>checkpoints</code>
<code>/home/user/workspace/**</code>	<code>gateway.pid</code> , <code>processes.json</code> ; <code>/opt/fromdonna/**</code>

Implementation phases

Phase	Deliverable	Notes
0. Contract	Manifest schema, path filters, R2 bucket	Team sign-off
1. R2 plumbing	Bucket + Worker binding; read/write helpers	No product behavior yet
2. Post-turn save	Async backup after successful turn	User reply first
3. Restore on provision	Create/re-provision hydrates from R2	Closes the real gap
4. Session resume fix	Oneshot actually continues session	Harness / Hermes
5. Hardening	Incremental workspace, metrics, size caps	After v1 works
6. Product tools → R2	Agent tools for explicit durable files	Completes path C

Repo touch points

Area	Change
cloudflare/gateway	R2 binding; after-turn save; restore in provision path
E2B-Template/harness	Optional pack endpoint; session continue fix
documentation/deployment/	Update memorymanagement from aspirational → contract
Hermes (vendored)	Reuse backup filters; avoid deep fork unless needed

Ops / security

- R2 credentials: Worker secrets only (or presigned, short TTL).
- Prefix isolation: users/{userId}/ only for that user.
- Logging: never log archive contents; log sizes + success/fail.
- Encryption at rest via R2 defaults; extra crypto only if required.

Open items for the team

1. Confirm **async save after reply** (recommended) vs block on R2.
2. Confirm **Worker-pull** vs **sandbox-push with presigned URLs**.
3. Max size per user / per turn (e.g. skip or truncate workspace over N MB).
4. Retain only **latest** snapshot vs keep last K versions.

Product decisions already locked

- Save frequency: **per successful turn**
- Scope: brain + sessions + workspace (full continuity feel)
- Product files: filesystem sync + tools→R2 (path C)
- Secrets: never on R2
- Day-to-day idle: E2B pause remains primary live mechanism

Source investigation grounded in FromDonna repo: [cloudflare/gateway](#), [E2B-Template/harness + template.ts](#), [documentation/deployment/memorymanagement.md](#), [Hermes hermes_cli/backup.py](#) and oneshot path.